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ScienceDirect
Journal of the European Ceramic Society xxx (2014) xxx–xxx
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High lithium ion conducting solid electrolytes based on NASICON $Li_{1+x}Al_xM_{2-x}(PO_4)_3$ materials (M = Ti, Ge and $0 \leq x \leq 0.5$)

K. Arbi *, W. Bucheli, R. Jiménez, J. Sanz

Instituto de Ciencia de Materiales de Madrid (CSIC), Sor Juana Inés de la Cruz 3, 28049 Cantoblanco, Madrid, Spain

Received 15 July 2014; received in revised form 19 November 2014; accepted 20 November 2014

Abstract

Fast ion conductors $Li_{1+x}Al_xTi_{2-x}(PO_4)_3$ (LATP) and $Li_{1+x}Al_xGe_{2-x}(PO_4)_3$ (LAGP) with $0 \leq x \leq 0.5$ have been successfully prepared by the solid state reaction method and characterized using X-ray diffraction (XRD), nuclear magnetic resonance (NMR) and impedance spectroscopy (IS) techniques. The structural analysis showed that the main crystalline phase detected in the XRD patterns of prepared samples displays the rhombohedral NASICON-type structure (space group R-3c). From the analysis of the ^{27}Al and ^{31}P MAS-NMR spectra, octahedra occupation and cation distribution have been investigated; from ^{7}Li MAS-NMR spectra, structural sites and local mobility of Li have been analyzed. Information about long-range lithium mobility has been deduced from the analysis of IS data recorded in the frequency 20 Hz to 3 GHz and temperature 100–500 K intervals. The use of the derivative $\log \sigma$ *vs.* $\log \omega$ function has enabled the detection of two high frequency responses that have been associated, according to the core-shell model, to the heterogeneous distribution of Al at surface and inside LATP and LAGP particles. IS measurements showed a higher bulk conductivity (3.4×10^{-3} and $\sim 10^{-4}$ S cm^{-1} at RT) and lower activation energy (0.28 and 0.38 eV) in LATP and LAGP samples, respectively.

Keywords: Fast ion conductors; NASICON; NMR spectroscopy; Electric impedance spectroscopy; Lithium batteries

1. Introduction

The necessity of modern society to develop sustainable transportation systems, reducing our dependence on fossil fuel, and the demand for a clean and secure energy source has stimulated a global resurgence in searching for advanced electrical energy storage and generation systems. A new generation of electric energy storage systems such as batteries, supercapacitors and fuel cells has been developed. Batteries remain the most promising electrical energy storage systems for many applications, from portable electronics to emerging technologies such as electric vehicles and smart grids. Among batteries, lithium ion batteries (LIBs) have attracted attention because the theoretical energy density (both gravimetric and volumetric) of lithium metal.

In the past decades, extensive investigations have been carried out on NASICON-type[1] lithium ion conductors with the general formula of $LiM_2(PO_4)_3$ (M = Ti, Ge, Hf, Zr, Sn).[2–10] Among them, $LiTi_2(PO_4)_3$ based materials seems to present the best characteristics for Li ion migration. When Titanium is partially substituted by trivalent N cation (N = Al, Ga, In, Fe, Sc, *etc.*) the "bulk" ionic conductivity is enhanced[11–16] reaching values comparables to those of organic electrolytes currently used in the rechargeable Li ion battery (10^{-2} Scm^{-1}).[16,17] In rhombohedral NASICON $LiM_2(PO_4)_3$ phases (S.G. R-3c), Li ions occupy mainly six-fold coordinated M1 sites, but in $Li_{1+x}R_xTi_{2-x}(PO_4)_3$ Li occupy M1 and M3 sites.[18] Unfortunately, the multivalence nature of the titanium can result in low dielectric stability caused by the reduction of Ti^{4+} *vs.* lithium in secondary lithium batteries. The investigation on alternative $Li_{1+x}Al_xGe_{2-x}(PO_4)_3$ (LAGP) materials[19–25] has recently gained considerable

Abbreviations: XRD, X-ray diffraction; MAS-NMR, magic angle spinning nuclear magnetic resonance; EIS, electric impedance spectroscopy; NASICON, Na-super ionic conductors; LATP/LAGP, lithium aluminum titanium/germanium phosphates.

* Corresponding author. Tel.: +34 913020440.
E-mail addresses: kamel@icmm.csic.es, kamel@ietcc.csic.es, rbikml@gmail.com (K. Arbi).

http://dx.doi.org/10.1016/j.jeurceramsoc.2014.11.023
0955-2219/

interest because they display a good electrochemical stability, a wide electrochemical window and have an ionic conductivity comparable to LATP compounds. LATP and LAGP materials are considered promising candidates for solid electrolytes; displaying good electrochemical performances in all-solid thin films $Li_4Ti_5O_{12}$/LATP/$LiMn_2O_4$ or Li/LAGP/TiO_2 LIBs devices operating at room temperature.[24,26]

In the present work we aim to investigate long-range mobility of lithium in $Li_{1+x}Al_xM_{2-x}(PO_4)_3$ systems (M = Ti, Ge and $0 \leq x \leq 0.5$) by impedance spectroscopy in wide temperature and frequency ranges. The MAS-NMR technique has been used to analyze cation location and Li mobility. This study will offer a better understanding of factors that enhance lithium conductivity in NASICON type materials.

2. Material and methods

LATP and LAGP samples, with $0 \leq x \leq x0.5$, were prepared by heating stoichiometric mixtures of Li_2CO_3, $(NH_4)_2HPO_4$, Al_2O_3 and TiO_2/GeO_2 in the interval 573–1373 K, following the procedure described previously.[15,17] All used reagents have a purity ≥99%. X-ray diffraction (XRD) patterns of NASICON samples were recorded with Cu-*K*α radiation (λ = 1.5405981 Å) in a PW 1050/25 Phillips apparatus to identify crystalline phases. Unit cell parameters of NASICON samples were determined with the Fullprof program (LeBail procedure).[27]

^{31}P, ^{27}Al and ^{7}Li MAS-NMR spectra were recorded at room temperature in a MSL-400 Bruker spectrometer (9.4 Tesla). The frequencies used for ^{31}P, ^{27}Al and ^{7}Li were 161.97, 100.4 and 155.50 MHz. ^{7}Li, ^{31}P NMR spectra were obtained after $\pi/2$ pulse irradiation (3 and 4 μs, respectively) and ^{27}Al NMR ones after $\pi/8$ (1 μs) single pulse irradiations, with a recycling time of 10 s. During spectra recording, the sample was spun at 2 kHz (^{7}Li signal) and 4 kHz (^{31}P and ^{27}Al signals) around an axis inclined $54°44'$ with respect to the external magnetic field (MAS technique). The number of scans was in the range of 100–800. ^{7}Li, ^{31}P and ^{27}Al NMR chemical shifts were referred to 1 M LiCl, 85% H_3PO_4 and 1 M $AlCl_3$ aqueous solutions. The fitting of NMR spectra was done with the Bruker WINFIT software package.[28] This program allows the position, linewidth and intensity of components to be determined with a non-linear least-square iterative method. The quadrupole C_Q and η parameters have to be deduced with a trial and error procedure.

To achieve the proper electrical characterization of samples two experimental set-ups were used: (i) an automatically controlled Agilent Precision LCR E4980-A analyzer for 20Hz–2 MHz and 77–550 K ranges. The sample was placed in a JANIS VPF 750 cryostat, allowing 4L-configuration measurements, (ii) an automatically controlled Agilent E4991-A RF Impedance/material analyzer for 1MHz–3 GHz and 150–550 K intervals. In conductivity determinations, the coaxial line was interrupted by inserting the sample between the central conductor of the Wiltron 18A50 air-line and the short circuit cap. Electrical conductivity measurements were carried out by complex Impedance Spectroscopy on 6 mm diameter and 0.5–1 mm thickness pellets. Samples were prepared by uniaxial pressing and sintered at 1273 K during 6 h.

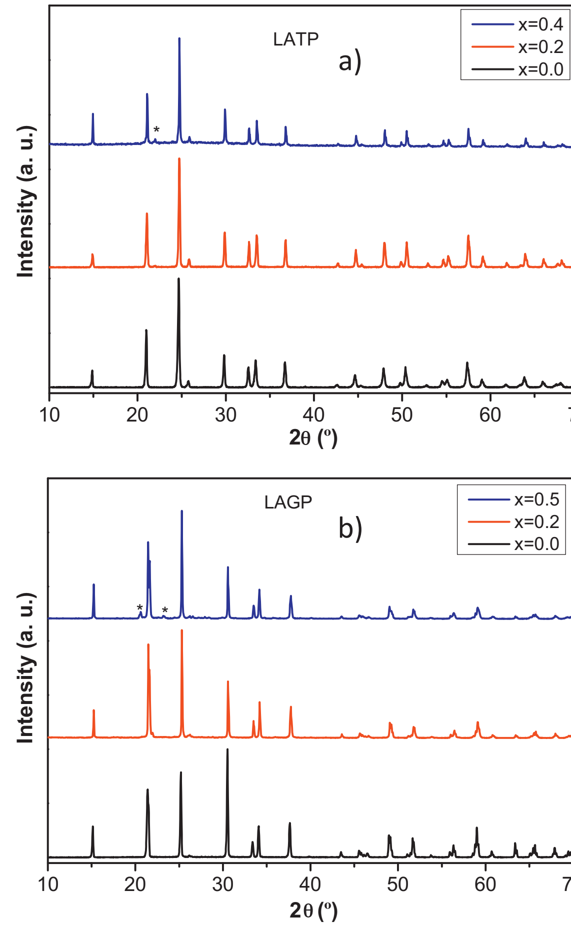

Fig. 1. XRD patterns of (a) LATP and (b) LAGP samples.

The larger surfaces of the pellets were polished until mirror finishing, then sputtered Pt electrodes were deposited. Finally, Pt Au paste electrodes (Dupont QG150) were painted and sintered at 1123 K during 1 h. After sintering, paste electrodes were finally polished. All the measurements were performed in vacuum or inert atmosphere. When necessary, non-linear least-square fittings of impedance data were performed with the ZView2 program.

3. Results

XRD patterns of analyzed samples annealed at 950 °C are shown in Fig. 1. The intense peaks of patterns correspond to those reported for $LiTi_2(PO_4)_3$ (JCPDS 35-0754) or $LiGe_2(PO_4)_3$ (JCPDS 41-0034) NASICON phases.[2,3] For $x > 0.2$, small peaks associated with aluminum phosphate were detected in both analyzed series. In LAGP samples; small amounts of GeO_2, Li_2O or $Li_4P_2O_7$ were also detected in Al-rich samples ($x = 0.5$). In LATP samples, XRD peaks shift toward higher 2θ positions and hexagonal a and c parameters decrease (Table 1) as a consequence of the smaller Al^{3+}size (0.535 Å) compared to Ti^{4+} cations (0.605 Å). In LAGP samples peaks do not shift appreciably because similar radius of Ge^{4+} (0.530 Å) and Al^{3+} (0.535 Å)[29] cations.

Table 1
Unit cell parameters of LATP and LAGP samples.

LATP			LAGP		
x	*a*	*c*	*x*	*a*	*c*
0	8.5112	20.8479	0	8.2771	20.4580
0.2	8.5020	20.8180	0.2	8.2830	20.4710
0.4	8.4975	20.7921	0.5	8.2828	20.5014

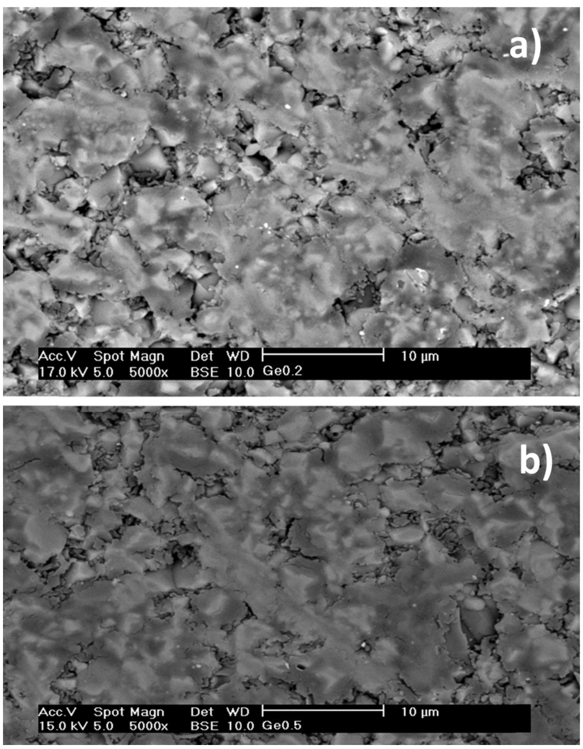

Fig. 2. SEM micrographs of the fracture surface of (a) LAGP-02 and (b) LAGP-05 ceramics.

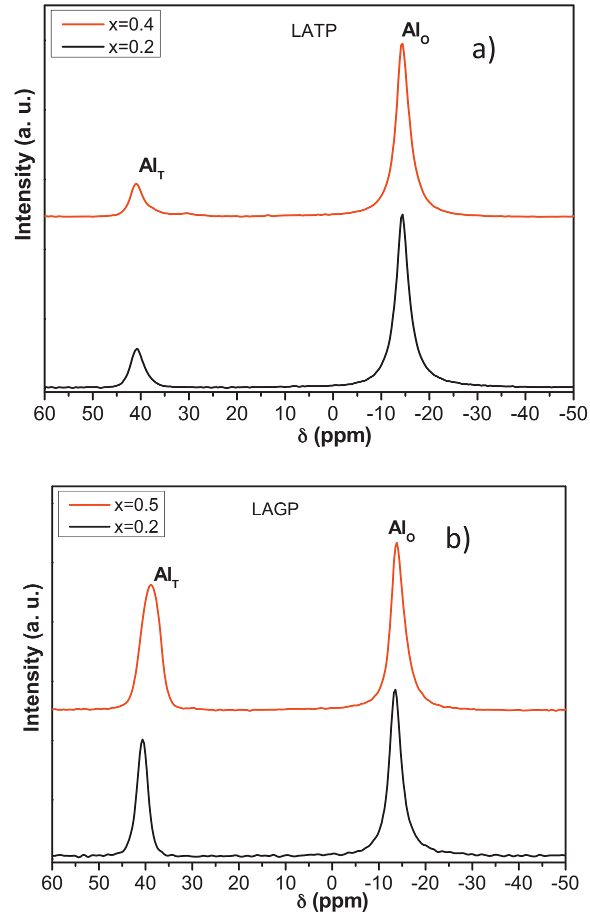

Fig. 3. ^{27}Al MAS-NMR spectra of (a) LATP and (b) LAGP samples.

Fig. 2 shows the fracture surface observed by scanning electron microscope (SEM) for LAGP ceramics. It can be observed that prepared ceramics have a very dense structure. The relative density of the pellets (deduced from the ratio d_{exp}/d_{the} of experimental and theoretical density values) was better than 90%. As an example, for $x = 0.2$, relative density values of LATP and LAGP are 92.8% and 90.1%, respectively. In some regions of SEM micrographs, porosity seems to be larger than 10%. This could be related to the polishing of the samples producing some pull-out and increasing the porosity in the polished surface of the analyzed ceramic.

^{27}Al MAS-NMR spectra ($I = 5/2$) of analyzed samples are formed by five ($-5/2 \rightarrow -3/2$, $-3/2 \rightarrow -1/2$; $-1/2 \rightarrow 1/2$ $1/2 \rightarrow 3/2$ and $3/2 \rightarrow 5/2$) transitions modulated by equally spaced sidebands associated to the sample spinning. The analysis of the central ($-1/2 \rightarrow 1/2$) transition in LATP and LAGP series showed one intense component at $\delta \approx -14$ ppm that corresponds to aluminum in octahedral coordination (Al_O) and one less intense at $\delta \approx 41$ ppm ascribed to tetrahedral coordination (Al_T), see Fig. 3. In LATP samples, the intensity of the Al_T signal is lower than that of Al_O one; however in LAGP samples, the intensity of the Al_T component is comparable to that of Al_O aluminum, Fig. 3b. The presence of tetrahedral Al has been associated with the presence of the secondary $AlPO_4$ phases. The substitution of Ti^{4+} by Al^{3+} in octahedra increases the amount of lithium in analyzed series.

^{31}P ($I = 1/2$) MAS-NMR spectra of LATP and LAGP samples are shown in Fig. 4. Spectra of undoped $LiTi_2(PO_4)_3$ and $LiGe_2(PO_4)_3$ samples display a single narrow line at −27.5 ppm and −43 ppm, ascribed to $P(OTi)_4$ and $P(OGe)_4$ environments. In LATP samples, the substitution of Ti^{4+} by Al^{3+} shifts toward more positive values and asymmetrically broadens the NMR line, indicating the presence of additional components. In the ^{31}P MAS NMR spectra of LAGP samples, additional signals were detected below −40 ppm, that have been ascribed to $P(OGe)_3(OAl)_1$, $P(OGe)_2(OAl)_2$, $P(OGe)_1(OAl)_3$, environments The separation between components increases with relative power strength of Al and Ti/Ge cations.

If it is assumned that Ti,Al and Ge,Al cation distributions are random in NASICON's compounds, it is posible to deduced Al/Ti y Al/Ge ratios with the expression[15]:

$$\frac{Al^{3+}}{M^{4+}} = \frac{4I_4 + 3I_3 + 2I_2 + I_1}{I_3 + 2I_2 + 3I_1 + 4I_0} = \frac{x}{2-x}$$

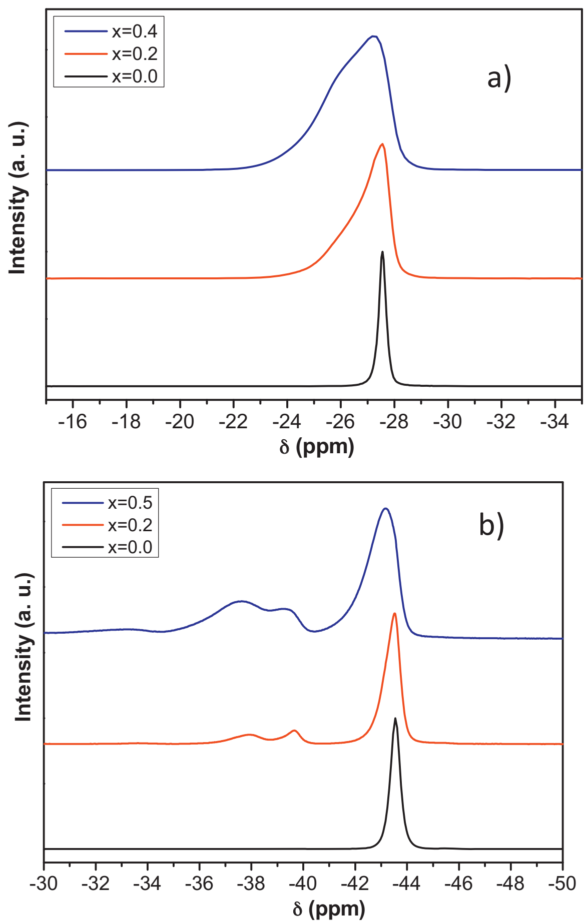

Fig. 4. ^{31}P MAS-NMR spectra of (a) LATP and (b) LAGP samples.

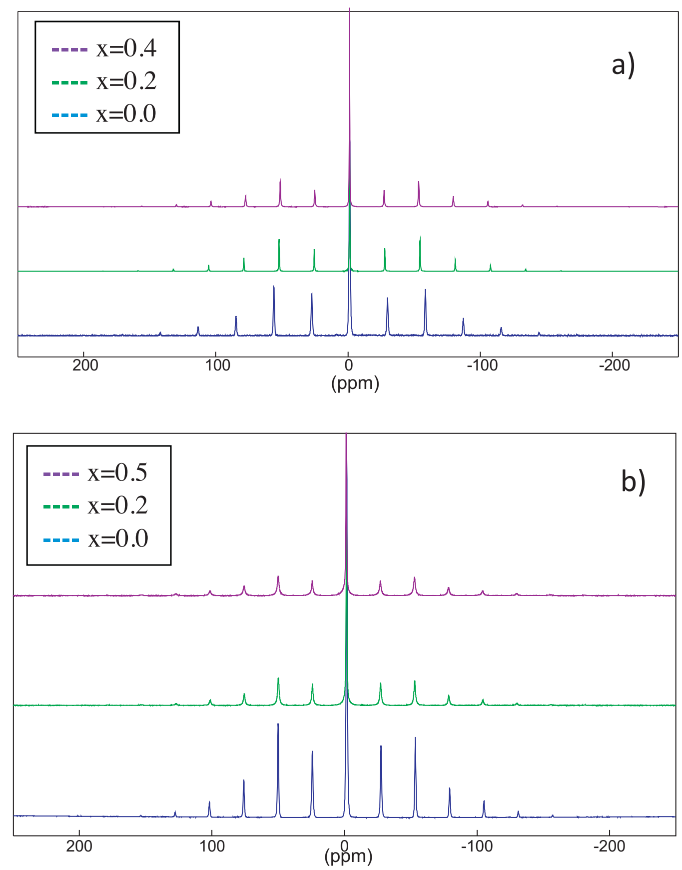

Fig. 5. ^{7}Li MAS-NMR spectra of (a) LATP and (b) LAGP samples.

Table 2
Experimental Al content incorporated in NASICON as deduced from quantitative analysis of ^{27}Al and ^{31}P NMR spectra.

LATP			LAGP		
x	$x_{exp}(^{27}Al)$	$x_{exp}(^{31}P)$	x	$x_{exp}(^{27}Al)$	$x_{exp}(^{31}P)$
0.2	0.167	0.186	0.2	0.129	0.164
0.4	0.334	0.348	0.5	0.280	0.317

where I_n stands for relative intensities of $P(OGe)_{4-n}(OAl)_n$ components and x and $2-x$ are relative occupancy of octahedral sites by Al^{3+} and M^{4+} (Ti^{4+} or Ge^{4+}) cations.

A comparison of Al contents deduced from ^{27}Al and ^{31}P MAS-NMR spectra are given in Table 2. In general values deduced from two NMR signals are similar, but appreciably different from nominal values, indicating that samples are heterogeneous and some segregation of Al phosphates is favored for high × values. At this point, it is important to deduce true Al values that optimize transport properties in NASICON phases.

^{7}Li ($I = 3/2$) MAS-NMR spectra of LATP and LAGP samples are formed by three possible (-3/2->-1/2; -1/2->1/2 and 1/2->3/2) transitions modulated by equally spaced sidebands associated to the sample spinning (Fig. 5). Spectra of these samples are formed by two non resolved components centered at ~0 ppm, ascribed to Li ions located at M_1 and M_3 sites. To better analyze both signals, satellite transitions were analyzed by enlarging the explored spectral region and using low spinning rates. From the fitting of experimental profiles, quadrupolar $C_Q \approx 25$ kHz and $\eta \approx 0$ parameters were deduced for Li ions at ternary axes (M1 sites). In the case of Li ions at M3 sites, only the central transition was detected, indicating that local mobility cancels Li-network interactions. The study of two signals suggests that exchange motion between two sites is still limited. From this fact, the quantitative analysis of Li MAS-NMR spectra based on simulation of experimental profiles, allowed the estimation of the relative occupation of two sites by lithium.

In fast ion conductors, the enlargement of the frequency window is required. The frequency range was extended by using two experimental set-ups described in the experimental section. The frequency dependence of electrical conductivity of LATP and LAGP series is depicted in Fig. 6. The analysis of conductivity plots at increasing temperatures allowed the resolution of "bulk" (B), "grain-boundary" (GB) and "electrode" (E) contributions. The contributions "B" and "GB" were better resolved applying the derivative criterion.[30] The LATP displayed a higher "bulk" conductivity than LAGP series: $\log \sigma_{B,300K} \sim -2.47$ for LATP and -4.1 for LAGP samples (Fig. 7a). The activation energy decreased from 0.28 eV to 0.24 eV as temperature increases in LTAP, but from 0.38 to 0.33 eV in LAGP samples.

The grain boundary response of ceramics samples are depicted in Fig. 7b. LTAP-02 and LGAP-02 samples presented the worst values for grain boundary conductivity:

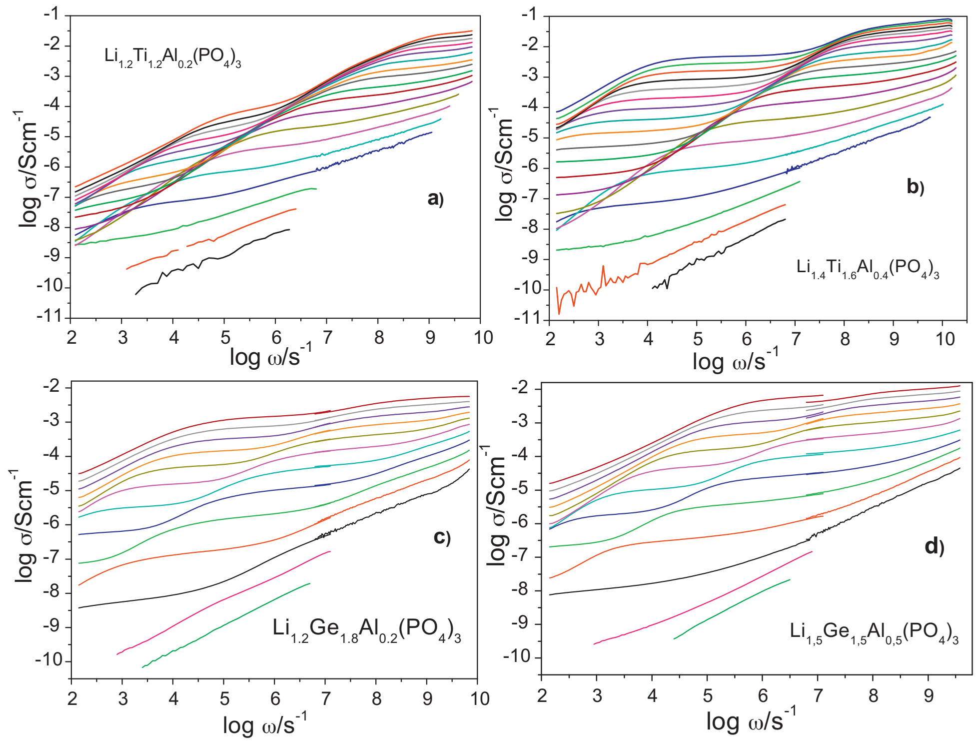

Fig. 6. Real conductivity *vs*. angular frequency at temperatures between 120 and 482 K with $\Delta T = 30$ K/Step for (a and b) LATP and (c and d) LAGP samples.

$\log \sigma_{GB,300K} = -5.92$ and -5.483, respectively. The samples with larger Al content presents similar values of $\log \sigma_{GB,300K}$: -5.05 and -4.91 for the LTAP-04 and LGAP-05. Activation energy for G.B. contributions were near 0.44 eV in both series.

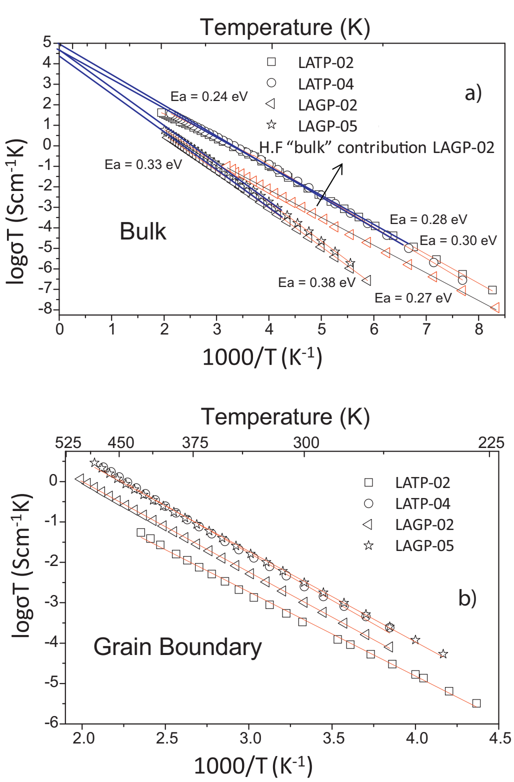

Fig. 7. (a) Bulk and (b) grain boundary dc-conductivities *vs*. reciprocal temperature in LATP and LAGP samples.

The total conductivity of LATP and LAGP samples become closer at RT.

4. Discussion

XRD patterns of prepared samples annealed at 950 °C agree with those reported for LATP/LAGP phases (Fig. 1). However, small reflections associated with $AlPO_4$ phosphates were detected in both series, indicating that nominal Al values were not obtained. In LAGP samples; residual amounts of GeO_2, Li_2O or $Li_4P_2O_7$ were also detected. The deconvolution of ^{27}Al and ^{31}P MAS-NMR spectra allowed a quantitative estimation of Al/Ti and Al/Ge ratios (Table 2). In general, it is observed that Al/Ti and Al/Ge are smaller than nominal values in NASICON compounds, and observed differences increase with the Al content. Moreover, the amount of Al incorporated in LAGP is lower than that of LATP series.

The substitution of tetravalent Ti^{4+} or Ge^{4+} cations by trivalent Al^{3+} cations increases the amount of Li in $Li_{1+x}Al_xTi_{2-x}(PO_4)_3$ (LATP) and $Li_{1+x}Al_xGe_{2-x}(PO_4)_3$ (LAGP) series. On the other hand, the fitting of experimental NMR profiles showed the presence of two signals ascribed to M1 and M3 sites. In this analysis, quadrupolar $C_Q \approx 25$ kHz and $\eta \approx 0$ constants were deduced for Li ions at ternary axes (M1 sites) and $C_Q \approx 0$ kHz and $\eta \approx 0$ for Li ions at M3 sites. In the second case, local mobility of lithium cancels Li-network interactions making that only the central component was detected.

The determination of Li1 and Li3 concentrations required of mathematical integration of satellite bands detected in MAS-NMR spectra. The analysis of LATP and LAGP series showed that the increment of Al content in NASICON phases favored the

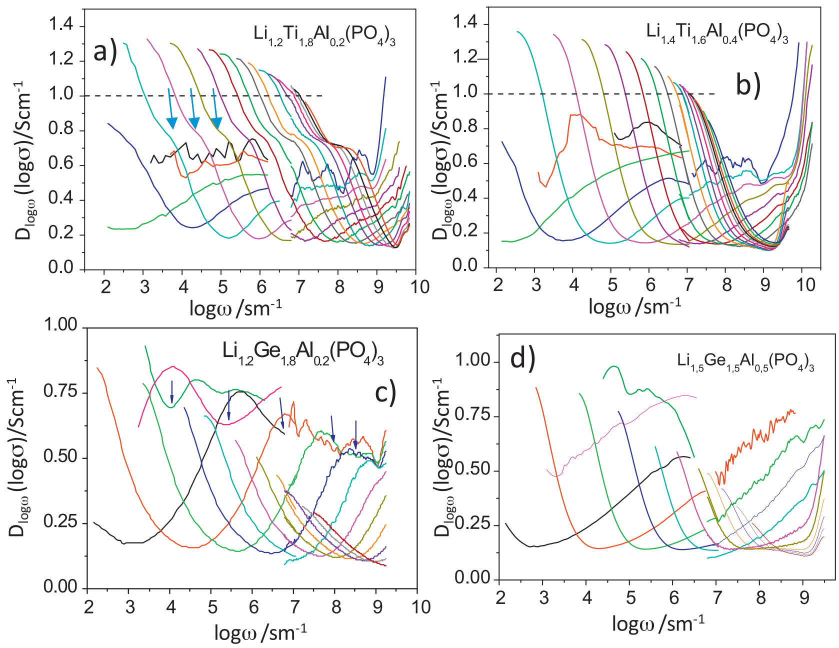

Fig. 8. Frequency dependence of derivative plots of (a and b) LATP and (c and d) LAGP samples. The temperatures are the same as in Fig. 6.

occupation of M3 sites, decreasing that located at M1 sites. This observation suggests that the occupation of M3 sites increases electrostatic repulsions between Li1 and Li3 ions, making Li1 ions to move to near vacant M3 sites ($d_{\text{Li–Li}} \sim 6.3$ Å).[18]

The temperature dependence of Li conductivity in two analyzed series is given in Fig. 7. In both series, conductivity plots display the Arrhenius behavior, showing some deviation from linearity at higher temperatures. LATP samples presented higher "bulk" conductivity than LAGP samples at RT: $\log\sigma_{300\text{K}} = -2.47$ for LATP and -4.1 S cm^{-1} for LAGP samples. In the temperature range explored, the activation energy of LTAP samples decreased from 0.28 eV to 0.24 eV, and from 0.38 to 0.33 eV in LAGP samples, when temperature increases. The decrease observed in activation energy has been previously ascribed to the creation of vacancy at M1 sites.[18] The formation of vacant sites at the intersection of conduction pathways, favors long-range mobility, in LATP and LAGP series.

A deeper analysis of conductivity revealed that the "bulk" response displays a higher degree of complexity. In this context, the "derivative criterion"[30] was used to resolve different contributions. In Fig. 8, plots of the derivative $\text{Dlog}(Y'(\omega))/\text{Dlog}(\omega)$ function are depicted at different temperatures for four analyzed samples. To better understand the expected shape of derivative functions, synthetic plots deduced from the fitting of admittance LAGP data to equivalent circuits[31] are given in Fig. 9a. The comparison of calculated and experimental plots showed that classical equivalent circuits did not reproduce the high frequency part of ceramic responses, suggesting the presence of an additional contribution.

In Fig. 8a, the derivative plot of conductivity is presented for the LATP_02 sample. At low temperature, a constant slope 0.65 was detected, that was related to the universal dielectric Jonscher response (UDR). The minimum detected at the intermediate frequencies, near the grain boundary contribution, is related to the "bulk" d.c. plateau (see arrows in figure), and that detected at the lowest frequencies and higher temperatures to the blocking effect at the grain boundaries. The strong unrealistic increment detected at GHz frequency in derivative plots was ascribed to the inappropriated compensation of the air line. The LATP-04

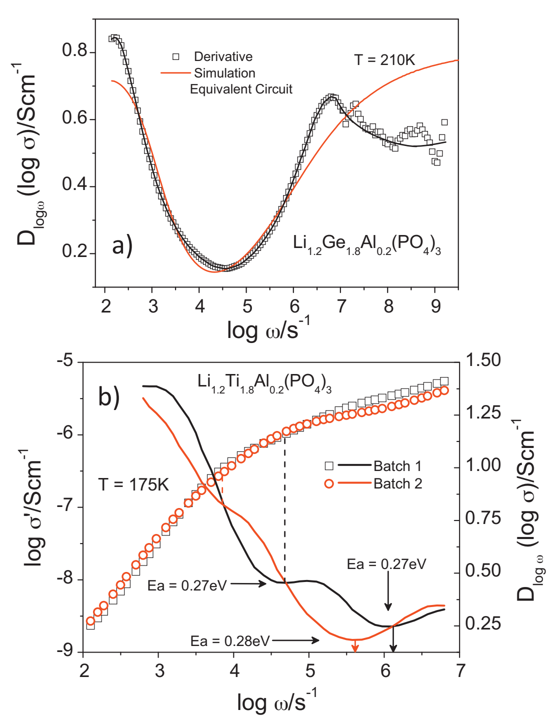

Fig. 9. (a) Derivative plots for LAGP-02 sample at 210 K. The red line stands for values calculated with equivalent circuits, and black squares are experimental values. (b) Conductivity *vs.* frequency plot for two batches of LATP-02 (red and black symbols). Derivative plots are given by red and black lines. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sample displays a single response (Fig. 8b) without described anomalies.

In the case of LAGP-02, however, an extra minimum was detected at the highest frequencies in dispersive region of the conductivity (denoted with arrows), Fig. 8c. The extra minimum indicates the presence of an extra dc-like-response. The presence of two "bulk" contributions has been reported previously in other NASICON phases, but this was not discussed.[17,32] The temperature dependence of the effective conductivity (Arrhenius plot) is given in Fig. 7a. In the case of the LAGP-02 sample the extra "bulk" high-frequency contribution displayed an activation energy of 0.28 eV that is close to that deduced for the main "bulk" response of LATP samples. The LATP-04 sample displays a single response (Fig. 7b) without described anomalies.

The presence of two "bulk" responses depends on preparation conditions of samples. To illustrate this effect, the real part of conductivity measured at 175 K for different batches of LATP-02 is given in Fig. 9b. The conductivity response is almost linear, with a small deviation in the high frequency range; however, derivative plots showed two contributions. In general, one single dc-"bulk" contribution is dominant in samples where derivative values at minima are close to zero. In one of analyzed batches, both contributions and their corresponding activation energies (0.28 eV) are similar.

In LATP samples analyzed in this work, the high-frequency "bulk" contribution dominates the "bulk" response, but this is reversed in LAGP samples. This effect can be related to compositional heterogeneities in NASICON particles, producing zones with different Al content. From ^{27}Al and ^{31}P MAS-NMR spectra analysis, the amount of Al incorporated in NASICON compounds is smaller than expected. In LATP and LAGP series, $AlPO_4$ impurities were detected for $x \geq 0.2$; this segregation was better detected in LAGP than in LATP samples (see Table 2). The segregation of Al at the surface particles depends on the composition and preparation conditions and strongly condition high-quality of ceramics. The Al depletion from the particles surface to form less conducting phases should have a strong influence on the total and grain-boundary resistances. In LATP-02 samples where the secondary contribution is displaced to much lower frequencies than the main "bulk" response, the grain-boundary blocking is much larger than in LATP-04, LAGP-02 and LAGP-05 samples.

The observed separation of "bulk" responses in LATP-02 and LAGP-02 samples can be explained with the core-shell model. According to this model, the relative importance of shell and core contributions should be different in LATP-02 and LAGP-02 samples. In the LAGP series, the low frequency "bulk" contribution displays a lower conductivity than the core contribution. In the case of the LATP sample the main "bulk" high frequency contribution, displaying high conductivity, suggests a higher volume for the core phase. This segregation must be related to the deficient Al incorporation into NASICON compounds. In LAGP samples, the Al incorporation to the NASICON phases is more difficult than in LATP samples, producing a thicker low Al/Li shell. In Fig. 10, a schematic view of the proposed core-shell model for two series is given. When Al increases, the segregation of $AlPO_4$ phase is favored in both series. This segregation

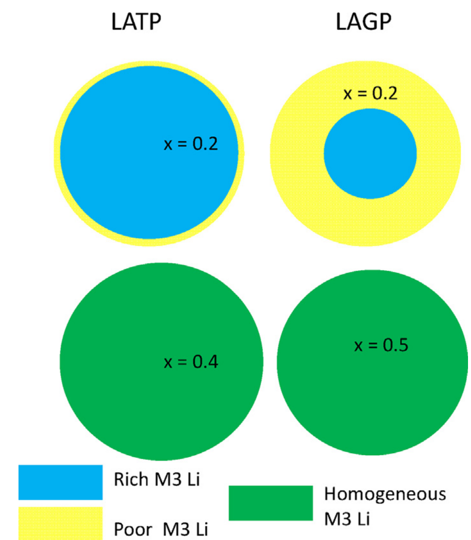

Fig. 10. Core–shell model used to explain observed electrical heterogeneity in NASICON samples. Yellow regions indicate poor-Al/Li phases and blue regions indicate rich Al/Li particles. Green regions stand for homogeneous materials. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

produces the partial elimination of the core-shell structure, favoring a more homogeneous composition in grains. This compositional homogeneity produces better grain boundary responses and improved ceramic Li conductivity.

It should be noted that from the obtained IS results it is possible to make a quantitative estimation of the relative fraction of the less Li conducting phase by using suitable composite models, joined to an in-deep micro-structural analysis of the ceramic samples. From deduced information the effective core-shell structure can be deduced. The results of this analysis will be the aim of a future work.

5. Conclusions

The analysis of LATP and LAGP series by XRD, NMR and IS has permitted to show that the increment of conductivity detected in Al-doped samples is associated with the Li content incorporation. In these samples, the increment of Li content improves electrostatic Li–Li repulsions between Li ions located at M1 and M3 sites, favoring the occupation of M3 sites and localization of vacant sites at M1 sites. In M3 sites, quadrupole interactions, measured at room temperature, are much lower than at M1 sites, indicating that Li-network interactions are reduced, increasing local Li mobility at M3 sites. Both facts favor long range mobility of lithium in conduction pathways.

Taken into account low Al concentrations in prepared samples, some heterogeneity has been detected in low Al contents, suggesting the existence of poor-Li shells structures, with an Al-rich core. On increasing the Al substitution the electrical response is homogeneous related to a homogeneous grain composition. The importance of compositional heterogeneities depends on the sample preparation. The derivative criterion used in conductivity data has demonstrated the existence of two well-resolved contributions in heterogeneous samples. In order to increase Li conductivity, the improvement of Al homogeneity in particles is required.

Acknowledgments

Authors thank Spanish Ministry of Economy (projects MAT2010-19837-C06-02) and the Regional Government of Madrid (project S-2009/PPQ 1626) for financial support.

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